LESSON 4.1 | Lesson notes

malloc and free

Overview

The function malloc asks the heap for a number of bytes and returns a pointer to them, or NULL if there is not enough memory. To allocate room for ten ints write int *a = malloc(10 * sizeof(int)); and check that a is not NULL before using it. When you have finished, call free(a); to return the memory. Every successful malloc needs exactly one free. Calloc works like malloc but also sets the memory to zero. Set a pointer to NULL after freeing it so it cannot be used by mistake.

Key points

- Part of: Dynamic Memory
- About 25 minutes
- Try the example yourself, then check it

How to use this handout

1. Read the overview first, then work through the key points in order.
2. Write down anything you are unsure about and ask in the course forum.
3. Use page 2 to plan your practice and keep notes.

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Practice checklist

- [ ] Read: malloc and free
- [ ] Try the example from the lesson
- [ ] Explain the main idea in your own words
- [ ] Answer the related questions
- [ ] Ask about anything unclear in the forum

My notes